A 1D-ARC full dataset

Figure 10: **1D ARC: visualizations of selected sample tasks.** For each task type, one pair of input-output images is shown.

Table 5: **Full results of direct-grid and object-based approach on the 1D-ARC dataset.** Each row displays results for one generated task type using different methods. The values correspond to the number of tasks out of 50; higher is better and the top-performer is bolded for each task type.

Task	Direct-grid		Object-based	
	GPT-3.5	GPT-4	GPT-3.5	GPT-4
Move 1	10	33	39	**50**
Move 2	3	13	22	**50**
Move 3	7	12	14	**49**
Move Dynamic	6	11	7	**37**
Move 2 Towards	3	17	17	**50**
Fill	6	33	44	**49**
Padded Fill	3	13	37	**44**
Hollow	2	28	40	**48**
Flip	11	35	20	**50**
Mirror	4	10	6	**13**
Denoise	11	18	48	**48**
Denoise Multicolor	13	30	36	**50**
Pattern Copy	11	18	31	**45**
Pattern Copy Multicolor	16	19	21	**47**
Recolor by Odd Even	13	**16**	15	13
Recolor by Size	2	14	21	**40**
Recolor by Size Comparison	6	10	17	**28**
Scaling	14	44	34	**46**

B 1D-ARC Design Process

1. **Identify ARC task that requires core prior knowledge.**

   See Figure 11(Left).

2. **Determine the solution and the inherent knowledge priors.**

   Solution: Recolor objects based on their sizes.

   Knowledge priors:

   - Objectness (Object cohesion)
   - Numbers and Counting priors (Understanding of object size through counting)

3. **Develop the 1D-ARC task using the identified solution and knowledge priors.**

   See Figure 11(Right).

4. **Reiterate steps 1-3 to create a comprehensive 1D-ARC dataset.**

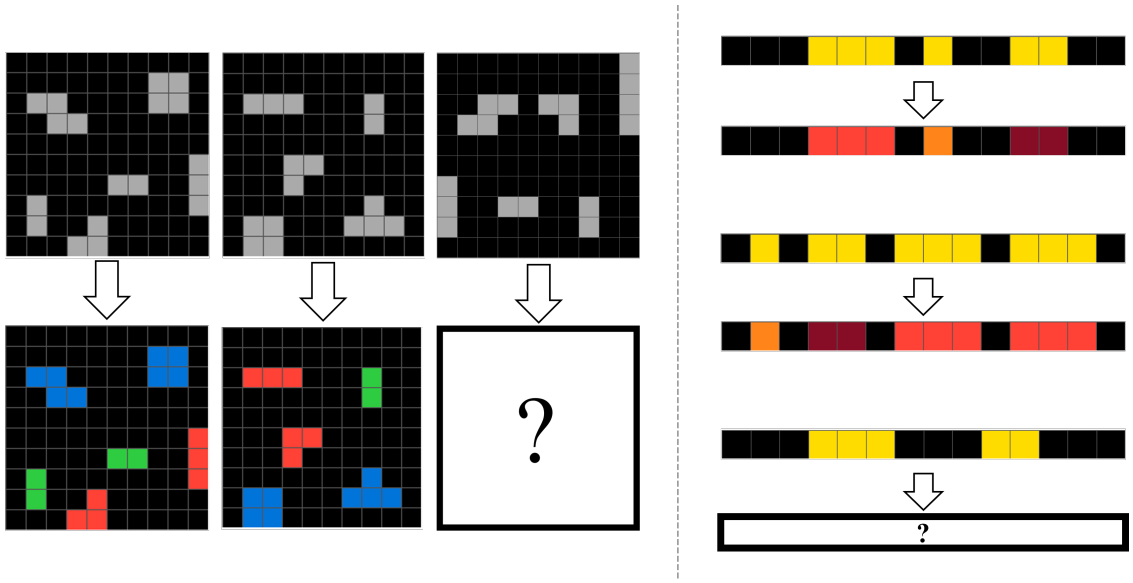

Figure 11: **Example of an ARC task (Left) and a 1D-ARC task adapted from it (Right)**